



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE

Batch : 2025-29

Assignment :

Subject: Programming with Java

ASSIGNMENT No :3

TITLE :Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.



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Code and Output Screenshots

```
J Overload.java > ...
1  public class Overload {
2      int add(int a,int b) {
3          return a + b;
4      }
5      int add(double a , double b) {
6          return (int)(a + b);
7      }
8      int add(int a , int b , int c) {
9          return a + b + c;
10     }
11
12     Run main | Debug main | Run | Debug
13     public static void main(String[] args) {
14         Overload calc = new Overload();
15
16         System.out.println("Addition of two integers: " + calc.add(10, 20));
17         System.out.println("Addition of two floating-point numbers: " + calc.add(10.5, 20.7));
18         System.out.println("Addition of three integers: " + calc.add(10, 20, 30));
19     }
20
21 }
22
```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Overload.java && java Overload
Addition of two integers: 30
Addition of two floating-point numbers: 31
Addition of three integers: 60

[Done] exited with code=0 in 0.567 seconds
```



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```
J Bank.java > Java > Bank > main(String[] args)
1  public static class Bank {
2      void interest() {
3          System.out.println("Interest is 5%");
4      }
5  }
6  public static class SavingsAccount extends Bank {
7      void interest() {
8          System.out.println("Interest is 7%");
9      }
10 }
11 public static class CurrentAccount extends Bank {
12     void interest() {
13         System.out.println("Interest is 3%");
14     }
15 }
Run main | Debug main | Run | Debug
16 public static void main(String[] args) {
17     System.out.println("Bank Interest Rate(Base):");
18     Bank b = new Bank();
19     b.interest();
20
21     System.out.println("\nSavings Account Interest Rate:");
22     SavingsAccount sa = new SavingsAccount();
23     sa.interest();
24
25     System.out.println("\nCurrent Account Interest Rate:");
26     CurrentAccount ca = new CurrentAccount();
27     ca.interest();
28 }
29
```

PROBLEMS 16

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, !excludeText, t...

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Bank.java && java Bank

Bank Interest Rate(Base):

Interest is 5%

Savings Account Interest Rate:

Interest is 7%

Current Account Interest Rate:

Interest is 3%

[Done] exited with code=0 in 0.693 seconds



Conclusion

This application demonstrates method overloading by allowing the same method name to perform operations with different parameter lists, improving flexibility and code readability. It also shows the use of static variables and methods, which are shared across all objects and can be accessed without creating an instance, making them useful for storing common data and utility functions.

Exercise

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

Ans)

```
src > AddNum.java > AddNum
1 public class AddNum {
2
3     int add(int a, int b) {
4         return a + b;
5     }
6
7     float add(float a, float b) {
8         return a + b;
9     }
10
11     Run | Debug
12     public static void main(String[] args) {
13
14         AddNum obj = new AddNum();
15
16         System.out.println("Addition of 11 and 52 is : " + obj.add(a: 11, b: 52));
17         System.out.println("Addition of 10.2 and 4.3 is : " + obj.add(a: 10.2f, b: 4.3f));
18     }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER Run: AddNum + - ...

Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Windows 11\Desktop\JAVA Programs(PT)-20260711T072533Z-2-001\JAVA Programs(PT)\AddNumNew.java> cmd /C ""C:\Program Files\Java\jdk-26.0.1\bin\java.exe" --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\Windows 11\Desktop\JAVA Programs(PT)-20260711T072533Z-2-001\JAVA Programs(PT)\AddNumNew.java\bin" AddNum "

Addition of 11 and 52 is : 63
Addition of 10.2 and 4.3 is : 14.5



2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Ans)

```
Restaurant.java > Java > Restaurant
1 public class Restaurant{
2     static int totalOrders = 0;
3     //Dine-in
4     void CalcBill(int itemPrice) {
5         totalOrders++;
6         System.out.println("Dine-in Bill: " + itemPrice);
7     }
8
9     //Takeaway
10    void CalcBill(int itemPrice, int packingcharge) {
11        totalOrders++;
12        int finalPrice = itemPrice + packingcharge;
13        System.out.println("Takeaway Bill after packing charge: " + finalPrice);
14    }
15
16    //Home Delivery
17    void CalcBill(int itemPrice, int packingcharge, int deliverycharge) {
18        totalOrders++;
19        int finalPrice = itemPrice + packingcharge + deliverycharge;
20        System.out.println("Home Delivery Bill after packing and delivery charge: " + finalPrice);
21    }
22
23    Run main | Debug main | Run | Debug
24    public static void main(String[] args) {
25        Restaurant restaurant = new Restaurant();
26        restaurant.CalcBill(670); // Dine-in
27        restaurant.CalcBill(670, 50); // Takeaway
28        restaurant.CalcBill(500, 50, 110); // Home Delivery
29
30        System.out.println("Total Orders: " + totalOrders);
31    }
32 }
```

PROBLEMS 16 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Restaurant.java && java Restaurant
Dine-in Bill: 670
Takeaway Bill after packing charge: 720
Home Delivery Bill after packing and delivery charge: 660
Total Orders: 3

[Done] exited with code=0 in 0.512 seconds
```



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